



A review on molecularly imprinted magnetic solid phase extraction emphasizing the analysis of antibiotics in complex matrices: Design, preparation, and application

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ABSTRACT

Magnetic solid phase extraction (MSPE) has been widely employed in the isolation and enrichment of antibiotics in complex matrices because it presents various unique advantages over traditional SPE including simple operation, fast extraction procedure, low cost and eco-friendliness. In recently years, magnetic molecularly imprinted nanoparticles (MMINs) containing highly specific recognition performance have been widely used to specific extraction of antibiotics under the format of MSPE. In this connection, recent advances of MMINs in the analysis of antibiotic residues are reviewed. Firstly, the common structure of MMINs including core-shell, core-satellite, cavity structure and nanotube/nanosheet-supported structure were introduced. After that, a main section covers the synthesis strategies of MMINs including one-step preparation and multi-steps preparation (precipitation polymerization, emulsion polymerization, suspension polymerization and surface imprinting technique) were illustrated. Furthermore, the applications of MMINs in the selective extraction of various antibiotic residues (sulfonamides, quinolones, tetracyclines, β -lactams, aminoglycosides and so on) in environmental and food samples were summarized. Finally, the existing problems and future prospects of MMINs were outlined in the final section of the review.

1. Introduction

Due to the highly antimicrobial activity, antibiotics have been widely used to prevent and treat various diseases and bacterial infections in human, animals and plants [1]. So far, the mainly employed antibiotics include sulfonamides (SAs), quinolones (QNs), macrolides (MLs), β -lactams (BLAs), tetracyclines (TCs), aminoglycosides (AGs), chloramphenicols (CAPs), lincosamides (LMs) and others [2]. Because of the indiscriminate application and overuse, antibiotics are inevitably introduced into the environment, such as sewage effluents, surface waters, drinking water, and soil [3]. The prevalence of antibiotics and their metabolites may induce the emergence of antibiotic resistance in bacteria, which becoming a major concern globally [4]. According to the conservative estimate, the number of deaths caused by antibiotic resistance worldwide each year is as high as 700,000 and it will rise to 10 million by 2050, even higher than the expected number of deaths from

cancer (8.2 million) [5]. The detrimental effects of antibiotics on human health can occur even when they are in trace concentrations. In this context, there is an urgent need to establish sensitive and accurate methods for the monitoring of antibiotic residues to provide a scientific basis for the management and supervision of antibiotics.

Currently, chromatographic techniques such as high performance liquid chromatography (HPLC), HPLC coupled with tandem mass spectrometry (HPLC-MS/MS) and gas chromatography coupled with mass spectrometry (GC-MS) have been extensively employed for antibiotic detection at trace levels in complex samples [6–8]. However, appropriate sample preparation techniques should be employed prior to chromatographic analysis so as to reduce the effect of sample matrices and increase detected sensitivity. So far, all kinds of pre-concentration techniques such as liquid-liquid extraction (LLE) [9,10], solid phase extraction (SPE) [11–13], in-tube solid phase microextraction (IT-SPME) [14,15], multiple monolithic fibers solid-phase

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microextraction (MMF-SPME) [16], dispersive solid-phase extraction (d-SPE) [17], magnetic solid-phase extraction (MSPE) [18,19] and stir-bar sorptive extraction (SBSE) [20] have been developed to isolate and capture of antibiotic residues. Among the existing techniques, MSPE is attractive because it presents superior merits such as simple operation, quick extraction procedure, low consumptions of sample and solvent.

MSPE employing magnetic or magnetizable nanoparticles as adsorbents was first introduced by Šafaříková in 1999 and subsequently developed by many researchers [21–23]. Typically, adsorption, magnetic separation and desorption are the main steps of MSPE procedure (Fig. 1) [24]. During MSPE, magnetic adsorbents are dispersed in sample solution to capture the target analytes under the auxiliary agitation (e.g., oscillation, vortexion and sonication) for an appropriate time. After the adsorption, the dispersed magnetic adsorbent can be rapidly separated from the sample solution by applying an external magnetic field. The adsorbents adsorbing the targets are re-dispersed in the eluting solvent to release the analytes. Subsequently, the adsorbents can be reused after washing. MSPE affords various unique advantages and have been used in many fields such as environmental monitoring [25], food testing [26] and biochemical analysis [27]. In MSPE, the core is the magnetic adsorbents because it strong affects the extraction performance. So far, several magnetic adsorbents such as $\text{Fe}_3\text{O}_4\text{-COOH@hydrogen-bonded organic frameworks}$ [28], magnetic porous polymer composite [29], magnetic covalent organic frameworks (COFs) [30] and magnetic molecularly imprinted polymers (MIPs) [19] have been reported to capture antibiotics. Among the various magnetic adsorbents, separation media with highly specific recognition performance is appealing because it can effectively reduce the influence of sample matrices.

Molecular imprinting is a useful technique for creating artificial recognition sites for specific selectivity towards target molecules [31, 32]. MIPs, which are complementary to the templates in shape, size and functional groups, is an example of three-dimensional network organic polymers with tailor-made specific binding sites prepared by means of molecular imprinting technique (MIT) [33]. Fig. 2 depicts the scheme of the molecular imprinting process. Typically, the templates and functional monomers are mixed to form a three-dimensional polymer network by multiple interactions (such as hydrogen bonding, dipole-dipole, and ionic interaction etc.) under the assistance of cross-linker agents. After the polymerization process, the templates are removed from the polymer using a solvent, and the specific tailor-made imprinted cavities are formed, which matched with the target molecules in specific shape, structure and functional groups [33,34]. Therefore, compared with traditional adsorption materials, MIPs exhibited unique features of structure predictability, recognition specificity and application universality in various fields. In the preparation of MIPs, template, functional monomer and cross-linker are the “three-elements of molecular imprinting”. In the pre-polymerization reaction, the functional

groups of the template should keep stable and carry out self-assembly process with functional monomer. Typically, functional monomer can affect the selectivity and accuracy of molecular recognition sites. The type and the amount of cross-linker can influence the number of recognition sites, the morphology, mechanical properties, selectivity and binding capacity of MIPs [35,36]. Moreover, due to the advantages of good physicochemical stability, highly specific selectivity and excellent reusability, the combination of MIPs with magnetic nanoparticles, namely magnetic molecularly imprinted nanoparticles (MMINs) can improve the extraction selectivity and reduce the impact of interfering compounds.

Based on the plentiful studies of MMINs in antibiotic analysis (since 2010), we propose to review the recent advances of MMINs/MSPE technique in the selective isolation and concentration of antibiotic residues. The classification, preparation routes of MMINs and its application in the analysis of antibiotic residues are reviewed in detail. At the same time, the existing problems and challenges are analyzed, and the prospects for the future development of MMINs are proposed.

2. Classification and preparation strategy of magnetic molecularly imprinted nanoparticles

2.1. Structures of magnetic molecularly imprinted nanoparticles

So far, various structures of MMINs have been reported so as to meet the requirements of extraction. Based on the morphology, the structures of MMINs can be classified into core-shell structure [37], core-satellite structure [38], cavity structure [39] and the nanotube/nanosheet supported MIPs (Fig. 3) [40].

2.1.1. Core-shell structure

The core-shell structure is the most widely used of MMINs. It consists of a core (magnetic phase) and a shell (one or several polymeric phase layers) [41]. Since Ansell and Mosbach first prepared the core-shell MMINs in 1998 [42], the core-shell structure has been widely applied because of the unique properties such as dual properties from the core and shell materials, large surface area, highly thermal, chemical and mechanical stability. Moreover, the vast majority of the templates distributes on the surface or locates in the proximity of the surface region of magnetic core, resulting in more complete removal of the template and easier recombination of target molecules with the recognition sites, which greatly improves the selectivity [43–45]. In general, the magnetic core of MMINs are mainly divided into single component metal nanoparticles (Fe, Co, Ni, etc.), metal alloy nanoparticles (FeCo, FeNi, FePd, CoNi, etc.), and metal oxide nanoparticles (Fe_3O_4 , $\gamma\text{-Fe}_2\text{O}_3$, Co_3O_4 , etc.), which process strong ferromagnetic or superparamagnetic properties. Among them, iron oxides, such as Fe_3O_4 , $\gamma\text{-Fe}_2\text{O}_3$, MFe_2O_4

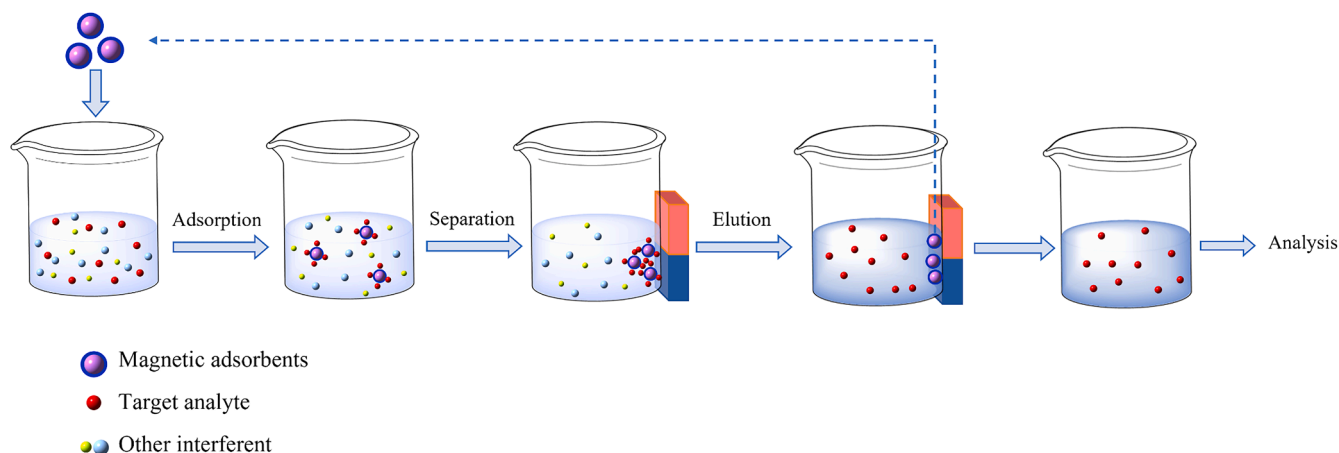


Fig. 1. The general scheme for MSPE procedure.

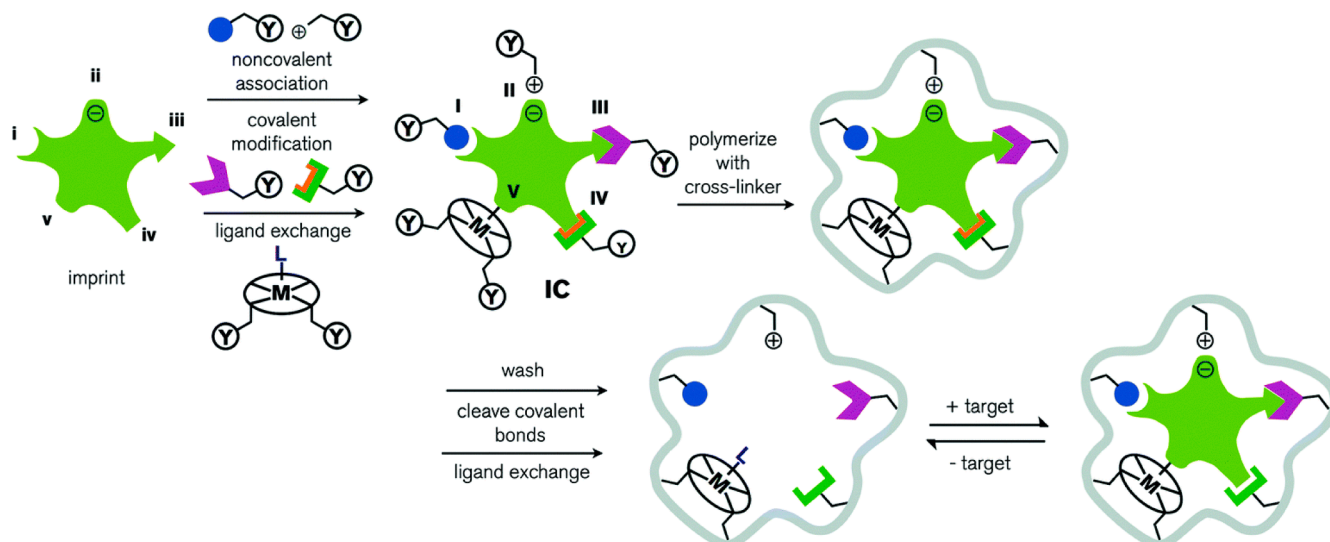


Fig. 2. Molecular imprinting process. (i) covalent; (ii) electrostatic/ionic; (iii) noncovalent; (iv) semi-covalent; (v) metal central coordination.

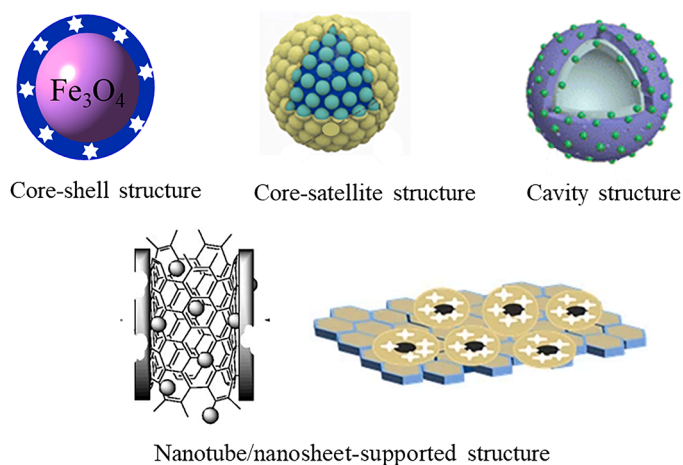


Fig. 3. The typical structures of MMNs.

(M=Cu, Mg, Mn, Co, Ni, etc.), have received considerable attention and become widely used as magnetic cores. Remarkably, the size of a magnetic material affects adsorption efficiency, and magnetic core with a diameter of 1–100 nm are preferably used in previous literature [46]. The magnetic core is produced by various synthesis approaches such as chemical coprecipitation, hydrothermal synthesis, sol-gel synthesis, solvothermal synthesis, pyrolysis and micro-emulsion method [47]. However, the bare magnetic cores obtained by the above-mentioned methods tend to agglomerate, resulting in low specific surface area and weak magnetism. Meanwhile, the bare magnetic cores easily decompose under acidic conditions and susceptible to oxidation or corrosion when it is exposed to air [48,49]. To overcome these limitations, the magnetic cores are coated with various materials (organic/inorganic materials). Silicon dioxide is one of the excellent coating materials for magnetic cores ($\text{Fe}_3\text{O}_4@ \text{SiO}_2$) due to its low price, chemical inertness, and thermal stability. Also, a silica network ($-\text{Si}-\text{O}-\text{Si}-$) formed via a condensation reaction after the hydrolysis of Si-O in tetraethoxysilane (TEOS) provides a hydrophilic surface, and are readily modified with MIPs without further modification in sol-gel polymerization [50,51]. Effting et al. synthesized $\text{Fe}_3\text{O}_4@ \text{SiO}_2@ \text{MIP}$ nanoparticles with core-shell structure and showed excellent selectivity and highly efficient extraction towards TCs in water and milk samples. In the study, monodisperse Fe_3O_4 were initially synthesized under facile

co-precipitation method, and subsequently coated with SiO_2 and oxytetracycline (OTC)-MIP layer to form the $\text{Fe}_3\text{O}_4@ \text{SiO}_2@ \text{MIP}$ nanoparticles in sol-gel polymerization and semi-covalent chemical imprinting approach, respectively [37]. In addition, Huang et al. used surface imprinting technique to synthesize magnetic MIP with a core-shell structure utilizing magnetite ($\text{C}@\text{Fe}_3\text{O}_4-\text{OH}$) as the magnetic core, trimethoprim as the template molecule, methacrylic acid (MAA) as the functional monomer. The resultant MMNs displayed a uniform morphology, good crystallinity, excellent magnetic properties and high binding performance with the maximum adsorption capacity of 137.4 mg/g [52].

2.1.2. Core-satellite structure

Core-satellite structured MMNs, which features a relatively large core (magnetic particles) surrounded with multiple nano-satellites, have obtained increasing interests as extraction medium for MSPE [53]. The core-satellite type MMNs exhibits better stability and adsorption performance than that of the conventional core-shell-structured MMNs. Moreover, one magnetic core of core-satellite structured MMNs can link a number of MIPs as the “satellites” replacing the cross-linked MIPs coating layer, which can ensure high enrichment capacity and fast transfer speed at the same time [54]. These features greatly favor extending the core-satellite structured MMNs to other field. Fan et al. prepared a type of core-satellite structured polydopamine-coated Fe_3O_4 nanoparticles-hollow porous MIP ($\text{Fe}_3\text{O}_4@ \text{PDA}-\text{HPMIPs}$) for the simultaneously selective enrichment of seven macrolide antibiotics residues [38]. The prepared $\text{Fe}_3\text{O}_4@ \text{PDA}-\text{HPMIPs}$ consisted of polydopamine-coated Fe_3O_4 nanoparticles ($\text{Fe}_3\text{O}_4@ \text{PDA}$) “core” linked to numerous hollow porous MIP (HPMIP) “satellites” with bridging amine functional groups. The as-prepared $\text{Fe}_3\text{O}_4@ \text{PDA}-\text{HPMIPs}$ revealed outstanding magnetic properties, high adsorption capacity (103.6 $\mu\text{mol/g}$), satisfactory selectivity and fast kinetic binding towards analytes.

2.1.3. Cavity structure

Cavity structure is gaining more and more attention due to its unique characteristics such as large specific and high stability. The cavity-structure MMNs can be divided into hollow-porous-supported MIPs (HP-MMNs) [55] and porous-materials-supported MIPs [56]. HP-MMNs refers to the cavity structures with pores on the surface and various hollow structure in the core, which contribute to the mass transfer of target analytes, and the high dense specific adsorption sites spreading at the internal and external surface of HP-MMNs can provide

superior recognition performance [55–58]. For example, Wang and coworkers prepared HP-MMINs for specific enrichment of trace tetracycline (TC) [55]. In this work, SiO_2 were synthesized as sacrificial support, small-sized Fe_3O_4 nanoparticles were synthesized as the magnetic source and MIPs were introduced on the internal surface of SiO_2 spheres. After the removal of TC and SiO_2 matrix, the sesame ball-like hollow structure was formed. The hollow structure provided large surface area and high dense recognition sites for adsorption of TC, which exhibited high adsorption capacity (70.23 mg/g) and ultrafast equilibration time (15 min) than traditional MIPs. Besides the aforementioned

structures, the structure of porous-supported MMINs was also reported. Li et al. prepared a magnetic porous cellulose MIP with Fe_3O_4 as the magnetic core and porous cellulose as the carrier, which exhibited high extraction capability (14.5 mg/g) and good selectivity (imprinting factors were in range of 1.46 to 3.14) than traditional MIPs [56]. A material with a similar structure was also fabricated and applied in the sensitive determination of trace norfloxacin (NFX) [59]. Even in complex samples, such as lake water and pork liver, the limit of detection (LOD) of the method was 6 ng/mL.

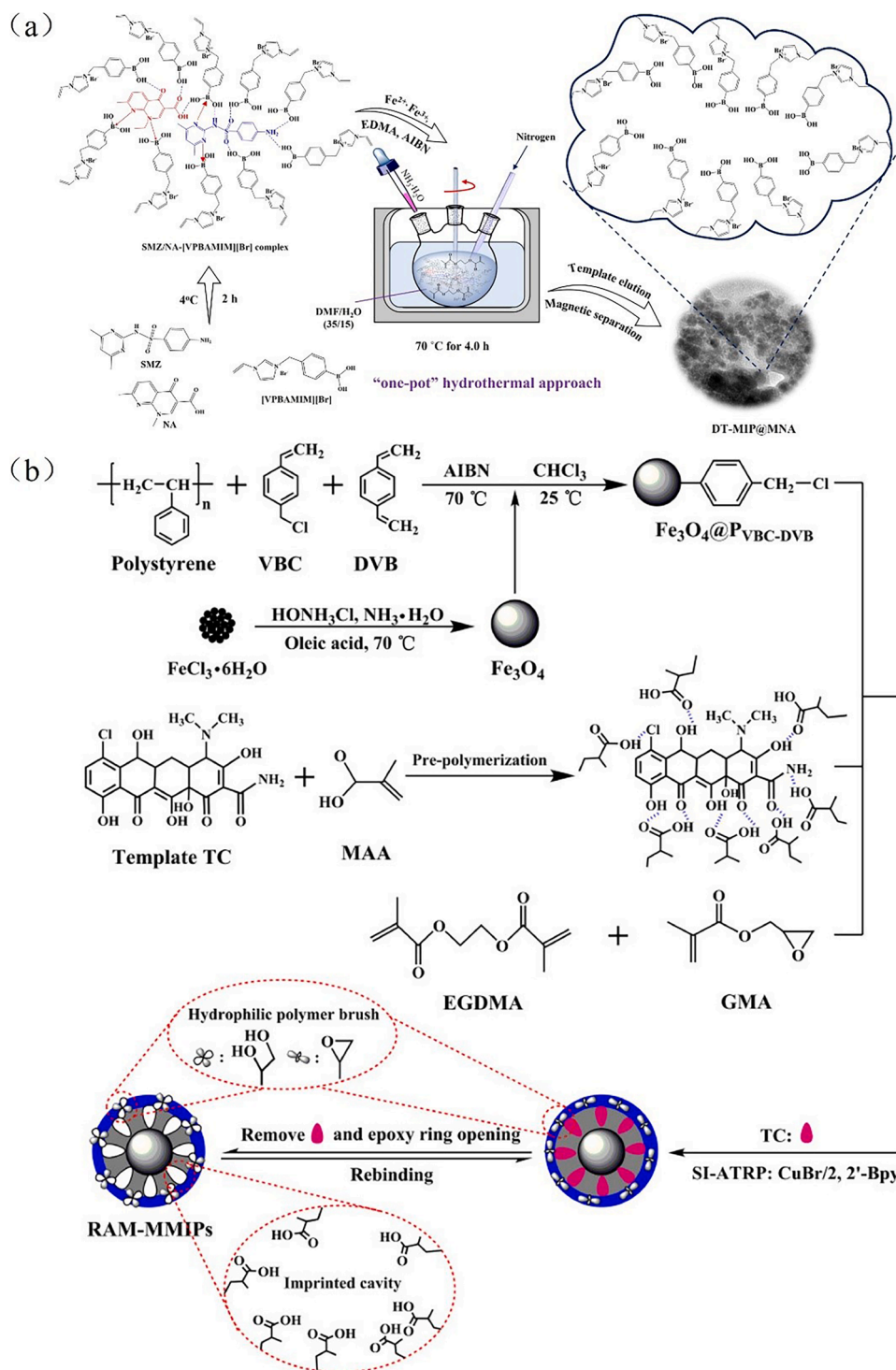


Fig. 4. Preparation of MMINs through (a) one-step strategy [19] and (b) multi-steps strategy [71].

2.1.4. Nanotube/nanosheet-supported structure

Typically, magnetic particles tend to attach on the surface of nanotube/nanosheet (such as carbon nanotubes, biochar, graphene oxide, and etc.) by co-precipitation or hydrothermal method. Incorporating nanotubes/nanosheets into the preparation of MMINs become promising candidates for preparing multifunctional MMINs, which display various advantages such as biocompatibility, high chemical and thermal stability, large surface area, specific selective sites and magnetic separation properties [60]. For instance, Han and Zhang et al. introduced magnetic metal-organic frameworks (MOF)/graphene oxide (GO) as the supporting materials to synthesize a nanosheet-supported structure MMINs (MOF/MGO/MIP) by using surface molecular imprinting technique for highly selective separation and extraction of SAs in food samples [39]. In the first step, the GO was prepared using an improved method. Secondly, GO combined with MOF (MIL-68(Al)/GO) was prepared via a solvothermal method. Subsequently, Fe₃O₄ was loaded on the surface of MIL-68(Al)/GO by co-precipitation. Finally, magnetic support (MIL-68(Al)/GO@Fe₃O₄), pre-polymerized solution containing sulfamethoxazole (SMX) and MAA, ethylene glycol dimethacrylate (EDMA) and 2,2-azoisobutyronitrile (AIBN) were mixed together to carried out sol-gel polymerization. The resultant MOF/MGO/MIP exhibited excellent binding capacity (42.65 mg/g) and specific selectivity (imprinting factor was 1.61) for SMX. In addition, Wang et al. synthesized MIPs on the surface of halloysite nanotubes (HNTs) for selective separation of OTC in environmental samples [61]. In this study, HNTs was modified with Fe₃O₄ via co-precipitation and the MIPs layer was imprinted on the surface of magnetic HNTs by the combination of surface molecular imprinting technology.

2.2. Preparation strategy of magnetic molecularly imprinted nanoparticles

The design of the MIPs layer on the magnetic core is of great importance during MMINs fabrication because it can influence the recognition capacity of MMINs. The primary factor in MIP imprinting polymerization is the strong bonding force between the template and the monomers. Therefore, choosing the appropriate functional monomer is a key factor in the design of MIPs. The application of computational simulation in designing MIPs has led to major advances in the selection of appropriate functional monomers, cross-linker, solvent and the monomer(s)-template ratio, which become the most extensively utilized approach in the design of MIP. Based on the preparation step, the strategy of MMINs synthesis can be divided into one-step and multi-steps (Fig. 4). In this review, the different strategies in the preparation of MMINs are outlined and discussed comprehensively. The typical preparation strategies of MMINs are summarized in Table 1.

2.2.1. One-step strategy

In one-step preparation (Fig. 4a), the precursor solution of magnetic particles and MIPs were mixed in the reaction vessel for integrating the in-situ growth of magnetic nanoparticles and reversed phase micro-emulsion oriented MIT [62]. The MMINs prepared by one step significantly simplifies the preparation process, improves the preparation efficiency and prevents a reduction in magnetism of magnetic core. Huang et al. used “one pot” synthesis strategy to synthesize a double-template MIPs (DT-MMINs) based on 1-vinyl-3-phenylboronic acid methyl-imidazole bromide ([VPBAMIM][Br]) as functional monomer and sulfamethazine (SMZ)/nalidixic acid (NA) as dual templates by one-step preparation strategy [19]. The average particle diameter of the magnetic core was 20 nm and the imprinting factor towards SMZ and NA were 5.89 and 2.90, respectively.

2.2.2. Multi-steps strategy

The typical process of multi-steps strategy of MMINs is showed in Fig. 4b. The corresponding procedures include the synthesis of magnetic core, surface modification of magnetic core, the introduction of MIPs layer and the removal of template molecule from the polymer [63]. In

Table 1

Comparison of different strategies of MMINs.

Imprinting methods	Advantages	Disadvantages
One-step strategy	Simple operation; one step preparation process	Consumption of large amount of template; template leakage; imprinting sites locate inside the polymer
Precipitation polymerization	One single preparation step; high-quality, uniform and spherical particles; no requirement of stabilizers; high yield	Consumption of large amount of template and porogen; high dilution factor; incomplete template removal; poor site accessibility; low binding capacities
Emulsion polymerization	High yield; monodispersed polymeric particles with small size (10–100 nm)	Suffers from the presence of surfactant remnants; removing surfactants is difficult; surfactant can affect the efficiency of polymerization; low imprinting capacity
Suspension polymerization	Simple process with one step polymerization; spherical particles	Big particle size; poor recognition
Surface imprinting	Higher utilization of binding sites; thin imprinted surface layer; monodispersed product; fast binding rate	A difficult and complicated process; time consuming

the preparation process of MMINs, MIPs layer endows MMINs with specific recognition ability towards target molecules. Therefore, the preparation of MIPs combining with magnetic core is the most critical step in the preparation process of MMINs. Free radical polymerization and sol-gel method are the commonly method used to modify magnetic core with MIPs [28]. Currently, free radical polymerization is the most widely used method for modifying magnetic cores with MIPs, which mainly occurs between the active groups (such as carbon-carbon double bonds) grafted on the surface of the magnetic core, monomers, and crosslinking agents [64]. Various synthesis strategies of free radical polymerization used by various researchers are precipitation polymerization [65–67], emulsion polymerization [59], suspension polymerization [68,69] and surface imprinting technique [18,70]. Some examples of the preparation of MMINs by free radical polymerization are listed in the Table 2.

Precipitation polymerization is also called the nonhomogeneous solution polymerization without stabilizer. In the process, templates, functional monomers, initiators and cross-linker agents are dissolved in porogen. After the polymerization, the resulting MIPs precipitate and encapsulate onto the magnetic core from the solvent. The reaction process displays advantages of simple, high-yield and does not require the addition of stabilizers to avoid complex subsequent processing. However, conventional polymerization processes require a large amount of porogen, accounting for about 95 % of the total reaction solvent volume [29]. Lopez et al. proposed an electrochemical sensor consisted of a carbon paste electrode modified with MMINs for selective recognition and quantification of amoxicillin and BLAs [65]. The MMINs were prepared by precipitation method via free radical using acrylamide (AM) as functional monomer, N,N'-methylenebisacrylamide as cross-linker and potassium persulfate as initiator. In the synthesis of ZnWO₄/γ-Fe₂O₃ MIPs by Yao et al., the ZnWO₄/γ-Fe₂O₃ was coated with MIP shells by precipitation polymerization method and then modified on glassy carbon electrodes to construct an electrochemical sensor for the determination of CAPs [66]. In addition, Reversible addition-fragmentation chain-transfer (RAFT) and atom transfer radical polymerization (ATRP) are the two types of controlled/living radical polymerization, which have been widely utilized to synthesize narrowly dispersed MIP beads, owing to their intrinsic advantages over traditional precipitation polymerization [67,71].

Oil phase containing MIPs precursor solution (template, cross-

Table 2

Overview on the synthesis strategies of MMINs by free radical polymerization towards antibiotics.

Adsorbents	Core	Templates	Monomers	Imprinted strategies	Synthetic steps	Reaction time (h)	Consumption of organic solvents (mL/g)	Ref
Fe ₃ O ₄ /SiO ₂ /MIP	Fe ₃ O ₄ /SiO ₂	OTC	3- (triethoxysilyl)propyl isocyanate	Semi-covalent method	4	52	>30.6	[37]
MOF/MGO@MIP	MIL-68(Al)/GO@ Fe ₃ O ₄	SMX	MAA	Surface imprinting method	5	72.5	1710	[39]
MIP	C@Fe ₃ O ₄ —OOC—CH=CH ₂	Trimethoprim	MAA	Surface imprinting method	3	51	206.5	[52]
magnetic MIPs	Fe ₃ O ₄	NFX	MAA	Emulsion method	3	23	141.5	[59]
Mag-MIP	Fe ₃ O ₄ @SiO ₂ -MPS	Amoxicillin	AM	Precipitation polymerization method	5	34	360	[65]
ZnWO ₄ /γ-Fe ₂ O ₃ /MIP	ZnWO ₄ /γ-Fe ₂ O ₃	CAP	α-MAA	Precipitation polymerization method	4	74	>901	[66]
PGr/ CdTe QDs/ Fe ₃ O ₄ @SiO ₂ /MIP probe	Fe ₃ O ₄ @SiO ₂	Cefoperazone	3- aminopropyltriethoxysilane	Suspension polymerization method	3	18.3	593	[68]
multi-templates MIP	MGO@mSiO ₂ and VTTS-MGO@mSiO ₂	FLRX,EN,NFX, CIP,ENRO, LOME	4- vinylbenzoic acid	Surface imprinting method	1	28.5	168	[70]
RAM-MMIPs	Fe ₃ O ₄ @PVBC-DVB	TC	MAA	SI-ATRP method	4	74	189	[71]
MIPs	hydrophobic Fe ₃ O ₄ nanoparticles	Erythromycin	MAA	Emulsion polymerization method	2	23	>2728	[73]
MMIPs	Fe ₃ O ₄ @SiO ₂	CFX	AM	Suspension polymerization method	3	61.75	2290	[74]
MUCPs@MIP	Fe ₃ O ₄	ENRO	MAA	“Layer-by-layer” assembly method	3	27.5	2192	[97]
Fe ₃ O ₄ @SiO ₂ @MPS	Fe ₃ O ₄ @SiO ₂	Doxycycline hyclate	MAA	Surface imprinting method	5	78	90	[113]
GQDs@ Fe ₃ O ₄ /MIP	Fe ₃ O ₄ @SiO ₂	Ceftazidime	3- aminopropyltriethoxysilane	Sol-gel polymerization method	3	18	676	[122]
MMIP Microspheres	FeO _x @SiO ₂ microspheres	Cloxacilin	MAA	Grafting polymerization method	4	31	377.6	[123]

linker), magnetic component and water phase containing surfactant are the basic composition of emulsion polymerization of MMINs. The preparation process involves transferring the oil phase and magnetic component into the water phase to get an emulsion, and initiating polymerization reaction under the action of initiator [72]. Through emulsion polymerization, small MMINs with a narrow size in nanoscale distribution can be generated. The prepared MMINs possess larger surface and can be widely used for imprinting water-soluble molecules. And the use of water medium can reduce the cost, improving the quality and safety. In conventional emulsion, discrete liquid droplets are stabilized by surfactants. Suitable amount of surfactant favors the formation of a stable core-shell system and a benign adsorption capacity of core-shell MIP. Zhang and Xie et al. synthesized a core-shell MMINs for NFX via oil-in-water emulsifier-free emulsion method [59]. The oil phase containing MAA-co-EDMA copolymer, Fe₃O₄ magnetic fluid and NFX (template) and water phase (4 mg/mL NaOH) were mixed to obtain MMINs. The core size was about 16 nm. However, the type of emulsion obtained using surfactants as stabilizers was not particularly stable. Pickering emulsion classified as either oil-in-water (O/W) or water-in-oil (W/O) polymerization has been used to prepare core-shell MMINs for selective adsorption/removal of antibiotics [73]. However, some shortages still can be encountered, such as the additional process should be employed to remove the residual surfactant on MMINs, which increases the complexity of the synthesis process.

Suspension polymerization is another commonly used method to prepare MMINs with a uniform size microspheres from small colloidal drops of the polymerization mixture suspended in a liquid phase. In this method, the MIPs precursor solution and magnetic core mixture are dispersed in a continuous phase through mechanical stirring, thus allowing a controlled precipitation of the MMINs into small size ranged from microgram to millimeter [64]. The particle size of MMINs prepared by suspension polymerization is influenced by mechanical stirring speed, type and quantity of surfactants, and continuous phase properties. Based on the component of continuous phase, suspension polymerization can be divided into conventional suspension polymerization and inverse suspension polymerization [72]. Chen et al. even reported the synthesis, characterization as well as the ability of MMINs to rapidly determine cephalixin (CFX) in complex animal-derived foods [74]. In this work, MMINs were synthesized via suspension polymerization using Fe₃O₄ magnetic nanoparticles as supporter, CFX, AM and EDMA as template, functional monomer and cross-linker, respectively. The adsorption capacity of MMINs toward CFX was as high as 24.18 mg/g and the established MMINs/MSPE-HPLC-UV method showed good applicability with excellent recovery (85.5 %–94.0 %) in determining CFX content. Li et al. described the synthesis of MMINs with double functional monomers by suspension polymerization, using chloramphenicol (CAP) as template and MAA/AM as double functional monomers [75]. The binding experiments revealed that the prepared MMINs

presented high adsorption capacity (42.6 mg/g) and strong recognition ability to CAP in food samples.

Another emerging method for the synthesis of MMINs is surface imprinting technique, forming uniformly recognition sites with high affinity in the outer layer or surface of the MIPs shell coated on the surface of magnetic carrier [76]. Due to the higher utilization of binding sites and good regeneration effect, surface imprinting technique has been proven to be an attractive preparation technique of MMINs. Hao et al. prepared a thermal-responsive magnetic MIP (TrMMIP) by surface imprinting technique. And then, a highly selective microextraction method based on TrMMIP/MSPE was developed for the recognition of SAs in complex food samples (milk, honey and fish) [18].

In addition, with the development of MIPs, researchers have proposed new strategies for preparing MMINs, such as living/controlled radical polymerization [1], electrochemical polymerization [29], surface grafting approaches [77] and dopamine self-polymerization [78]. The emergence of these new imprinting strategies provides wide opportunity for the application of MMINs of MMINs, and also expands the application fields of MMINs/MSPE technique.

3. Application of magnetic molecularly imprinted nanoparticles in the selective extraction of antibiotics

MMINs have demonstrated their enormous application prospect in the analysis of antibiotic residues, including SAs, QNs, MLs, BALs, TCs, AGs, CAPs, LMs and others. Table 3 summarized the application of MMINs in the analysis of antibiotics in various complex matrices.

3.1. Sulfonamides

SAs, a broad spectrum of antibiotics with a *p*-aminobenzene structure, are widely applied to treat bacterial infections in humans and animals [79]. Along with the uncontrolled or improper application, the residual SAs pose a risk to public health and ecological safety, resulting from their complex structures and low biodegradability [19,80]. Regarding the problem escalating, maximum residue limits of SAs in various foods have been regulated in some countries and organizations [80,84]. Based on the actual requirements, developing fast, sensitive and selective approaches to accurately monitor trace SAs residues in complex samples is desired.

As listed in Table 3, with the aid of MMINs, most of the analytical methods can achieve high sensitivity with low LODs at trace levels. Noticeably, certain pretreatment steps are needed prior to the extraction with MMINs to release the organic pollutants from samples. And liquid-solid extraction is used sometimes to dissolve the target analytes and transfer them into a solution so that MSPE can be carried out. So far, SAs residues in various samples including water, honey, milk, fish and shrimp have been successfully determined [18,19,39,40,81,82]. Normally, the quantification of SAs is realized by means of coupling with analytical instrument such as GC, HPLC, LC-MS, GC-MS and capillary electrophoresis. For example, Han et al. used SMX as template, adopting surface imprinting technique to synthesize MOF/MGO@MIP [39]. The synthesized MOF/MGO@MIP was used as the MSPE adsorbent to selectively extract four SAs in honey and milk samples, and then the analytes were detected with HPLC-UV. The results showed that the LODs were 0.39–0.47 µg/kg with an excellent linear correlation ($R^2 > 0.990$), and the spiked recovery were 81.38 %–96.71 %. The MOF/MGO@MIP also presented acceptable reusability and could be re-used at least seven adsorption-elution cycles. Huang et al. reported a polymer ionic liquid (PIL) modified MMINs through a quick and green one-pot strategy [19]. During the polymerization, SMZ (template), EDMA (cross-linker), [VPBAMIM][Br] (monomer) and $\text{Fe}^{3+}/\text{Fe}^{2+}$ (magnetic core) were mixed in porogen and reacted at 70 °C for 4.5 h. By the coupling of MSPE with HPLC equipped with diode array detector (DAD) analysis. The achieved LODs were 0.012–0.028 µg/L and 0.015–0.032 µg/kg for water and milk samples, respectively. Moreover, the extraction equilibrium was reached

within 8.0 min. Porous carbon derived from peanut shells was also used to prepare a magnetic porous carbon SAs MIP [52]. During the preparation process, utilizing peanut shells as carbon source, magnetic porous carbon material was successfully prepared and further modified into ethylene-based magnetic porous carbon through acryloyl chloride. The successful synthesis of MIPs was then achieved by employing surface imprinting techniques with ethylene-based magnetic porous carbon and MAA as bifunctional monomers. The obtained material was employed as the adsorbent of MSPE to selectively enrich SAs from actual samples prior to HPLC determination. The established method was applied to quantify four SAs in fish with high sensitivity, and the selective coefficients of MIPs towards target SAs compared to NIPs were in the range of 1.0–1.4 times. Moreover, the method not only exhibited high sensitivity but also presented remarkable accuracy and reliability. In addition, MMINs based on carbon nanotubes [40] and COFs [83] have been proposed as the medium of MSPE in selective separation and determination of SAs in complex samples.

3.2. Quinolones

Due to the powerful antibacterial activity against both gram-positive and gram-negative bacteria, QNs are widely applied to treat various diseases on humans and animals [87]. However, the widespread production and long-term use of QNs have led to potential residues in various samples such as water [70,88,89], soil [89], meat [90] milk [91, 92], honey [91], egg [89] and animal innards [93]. So far, various MMINs-based techniques have been developed for the selective pre-concentration of QNs in complex samples (Table 3).

As listed in Table 3, there are many studies on the detection and analysis of QNs in water samples based on MMINs/MSPE [70,89,91,94, 95]. Fan et al. prepared multi-templates MMINs via surface imprinting, using five quinolones, including fleroxacin (FLRX), enoxacin (EN), NFX, ciprofloxacin (CIP), enrofloxacin (ENRO) and lomefloxacin hydrochloride (LOME) as the templates, mesoporous silica modified magnetic graphene oxide as the carrier and 4-vinylbenzoic acid as functional monomer [70]. The developed MMINs/MSPE was combined with HPLC-DAD to detect trace QNs in water with the recoveries of 89.7–100.5 %. Dai et al. prepared a core-shell structure MMINs in water-containing systems for the extraction of QNs using heptakis (beta-cyclodextrin-ionic liquid) (ILs(2)-beta CD), ofloxacin (OFL), triallyl cyanate (TAC), and azobisisobutyronitrile as the functional monomer, template, cross-linker and initiator, respectively [94]. By means of hydrophobic and electrostatic interactions, the magnetic adsorbent exhibited good binding capacity (35.8 mg/g), special selectivity and excellent ability to eliminate matrix interference for QNs determination. Xie et al. developed an efficient approach to synthesize MMINs for selective extraction and detection of SAs and QNs in water samples [95]. In the study, 4-vinylbenzoic acid was selected as functional monomer, the imprinted layer was obtained on the magnetic graphene oxide modified with carbon-carbon double bonds. The obtained MMINs showed high adsorption capacities towards studied QNs (12.01–23.33 mg/g). Coupling with HPLC-DAD, the established method showed satisfactory recovery (81.49–100.7 %) and low LODs (0.012–0.015 µg/L) for QNs.

Because of high adsorption capacity and excellent selectivity, MMINs-based adsorbents are preferred to be used in the MSPE process for the analysis of QNs in various food samples. Adopting MAA-co-EDMA copolymer and NFX as monomer and template, respectively, Zhang and Xie et al. developed MMINs for direct enrichment of NFX from pork liver samples via oil-in-water emulsifier-free emulsion method [59]. The resultant MMINs not only showed uniform imprinting layers, stable crystalline phase and acceptable magnetism (3.052 emu/g), but also exhibited high adsorption capacity (27.04 mg/g) and fast adsorption kinetic. Studies about the monitoring of QNs residues in milk and dairy products also have been reported [19,96]. For example, Megia-Perez et al. employed Levofloxacin (LEVO), a non-regulated

Table 3
Applications of MMINs for antibiotic analysis.

Types of antibiotic	Templates	Monomers	Imprinted methods	Real samples	Analytical instruments	LODs (µg/L or µg/kg)	Recovery%	Ref.
SAs	Sulfadiazine	AM	Surface imprinting method	Milk, honey, fish	nESI-MS	0.2–1.0 µg/L	75.6–118	[18]
	SMZ	[VPBAMIM][Br]	One-pot method	Water, milk	HPLC-DAD	0.012–0.032 µg/L	74.3–120	[19]
	SMX	MAA	Surface imprinting method	Honey, milk	HPLC-UV	0.39–0.47 µg/kg	81.4–96.7	[39]
	Sulfabenzamide	MAA	/	Fish, shrimp	UPLC-MS/MS	0.1 µg/kg	90.2–99.9	[40]
	Trimethoprim	MAA	Surface imprinting method	Fish	UPLC-MS/MS	0.015 µg/kg	78.6–101	[52]
	SMX	MAA	/	Honey, milk	HPLC-UV	6.04 µg/L	68.3–80.1	[81]
	Sulfamethazine, sulfadiazine, sulfadimethoxine	MAA	/	Milk, water	HPLC-DAD	2.0 µg/L	84.3–96.2	[84]
	SMX	(3-Aminopropyl) triethoxysilane	/	Water	HPLC-UV	2 µg/L	94.2–98.2	[85]
	SMZ, SMX	MAA	/	Milk	HPLC-UV	0.03–0.06 µg/L	87.0–98.5	[86]
	FLRX, EN, NFX, CIP, ENRO, LOME	4-vinylbenzoic acid	Surface imprinting method	Water	HPLC-DAD	0.011–0.015 µg/L	89.7–101	[70]
QNs	OFL	ILs(2)-beta CD	Surface imprinting method	Water	HPLC-UV/FLD	0.43–1.83 µg/L	80.1–107	[94]
	ENRO, OFL, LOME, GATI	4-vinylbenzoic acid	/	Water	HPLC-DAD	0.012–0.015 µg/L	81.5–100	[95]
	NFX	MAA	Emulsifier-free emulsion method	Pork liver	HPLC-UV	6 ng/mL	77.4–84.7	[59]
	ENRO	MAA	"Layer-by-layer" assembly method	Fish	Fluorescence spectrum	0.406–1.12 pmol/L	90.3–108	[97]
	OTC	N-isopropylacrylamide	/	Tea	Fluorescence detection	0.1 µg/L	94.2–105	[104]
TCs	TC	MAA	/	Water, beef, milk	HPLC-DAD	91–296 µg/L	93.9–113	[56]
	Doxycycline	N-isopropylacrylamide	/	Water	Fluorescence detection	0.2 µg/L	92.5–105	[105]
	TC	Aptamers and β-cyclodextrin	One-pot method	Fish feed	HPLC-SPD	1.0 µg/L	85.0–112	[112]
	Doxycycline	MAA	Surface imprinting method	Sediment	HPLC-DAD	0.03 µg/g	90.6–93.8	[113]
	TC	MAA	/	Water	HPLC-DAD	3.75 µg/L	91.2–101.3	[114]
	TC	MAA	Surface imprinting method	Chicken liver	HPLC-PDA	0.0073–0.024 mg/kg	89.3–97.1	[115]
	TC	MAA	/	Milk	HPLC-UV	0.025–0.083 mg/L	86.6–100	[116]
	TC	MAA	ATRP	Milk egg	HPLC-UV	1.03–2.67 µg/L	84.2–98.6	[71]
	PENV	MAA	/	Milk	HPLC-MS/MS	1.6–2.8 µg/L	71.6–90.7	[119]
	CFX	AM	/	Milk, egg, beef	MS	0.3–0.5 µg/kg	72.6–115	[120]
BLAs	Cefoperazone	3-aminopropyltriethoxysilane	Suspension polymerization	Milk	Fluorescence measurement	0.09 µg/L	85.3–98.3	[68]
	Amoxicillin	AM	Precipitation method	Lake, milk	Electrochemical detect	0.75–2.48 µmol/L	96.7–100	[65]
	Ceftazidime	3-aminopropyltriethoxysilane	Sol-gel polymerization	Milk	Optosensor	0.05 µg/L	90.7–99.2	[122]
	Cloxacilin	MAA	Grafting polymerization	Pig plasma	Raman Spectroscopy	7.8 pmol	85.0–126	[123]
	Spiramycin	MAA	Sol-gel polymerization	Honey	HPLC-MS/MS	0.003–0.075 µg/kg	84.2–117	[124]
MLs	Erythromycin, azithromycin	MAA	Sol-gel polymerization	Pork, chicken	Piezoelectric sensor	0.5 µg/mL	101–128	[126]
	Raffinose	MAA	Surface imprinting method	Milk	HPLC-MS/MS	3.6–9.6 µg/kg	82.6–114	[127]
AGs	Kanamycin, tobramycin, gentamicin	TEOS	Suspension polymerization	Milk	LC-MS	1.07–4.88 nmol/L	95.4–105	[69]
	CAP	α-MAA	Precipitation polymerization	Milk	Electrochemical detect	0.0027 µmol/L	96.8–98.9	[129]
CAPs	CAP	MAA	Microwave-assisted method	Meet, honey, milk	Raman Spectroscopy	0.004 nmol/L	95.2–107	[130]
	Thiamphenicol	O-phenylenediamine	/	Milk, honey, meat	Cyclic Voltammetry	0.03 µmol/L	95.1–105	[131]
	CAP	MAA	Surface imprinting method	Marine sediments	HPLC-DAD	0.1 g/L	77.9–103	[132]
	Lincomycin hydrochloride	4-vinyl pyridine	ATRP	Milk	HPLC-UV	15.0 ng/g	94.3–98.2	[133]

quinolone, as template to prepare MMINS for selective capture of ENRO, CIP, sarafloxacin and NFX [96]. The resulting material showed high extraction selectivity towards ENRO and CIP in milk samples. In addition, a fluorescent probe prepared by means of "layer-by-layer" assembly of MIP and Fe_3O_4 nanoparticles on NaYF_4 were used to determine five QNs in fish tissues and the recoveries ranged from 90.33 % to 108.43 %. This work offers a new method based on fluorescence probe with magnetic and molecular recognition capability for rapid and accurate sensing of multiple chemical residues in environment and agri-food products [97].

3.3. Tetracyclines

TCs are antibiotics widely used to prevent and treat diseases in farm animals and as additives to promote growth due to their spectral antimicrobial activity and low cost. However, the overuse of TCs through contaminated food and water can cause allergic reactions in susceptible populations and consequently resistance to the drugs [98,99]. Due to the low concentrations of TCs residues and the complex sample matrices, sample preparation is indispensable for the accurate and sensitive determination of TCs. To selectively capture TCs, various MMINS have been designed and prepared.

Anaya-Castro et al. used magnetite (Fe_3O_4) as the magnetic core, TC as the template, 3-aminopropyltriethoxysilane as the functional monomer and TEOS as the cross-linker to synthesize novel superparamagnetic hydrophilic MMINS (MNP@ SiO_2 -MIP) for the removal of TC [100]. The obtained adsorbent exhibited good adsorption capacity with 23 mg/g, significant selectivity (imprinting factor of 2.2) against other molecules with similar functional groups and size, and the achieved removal rate was 80.1 %. Recently, a sensor based on MMINS was prepared for the monitoring of TC residue in food samples by Zeb et al. [101]. The prepared MMINS offered high enrichment efficiency and adsorption capacity, and the maximum adsorption capacity was as high as 8.25 mg/g. Moreover, the adsorption of TC on MMINS followed the Langmuir adsorption isotherm model and pseudo-second-order reaction kinetics. The developed method was successfully applied to determine TC in honey, cow milk and egg samples, with satisfactory LOD (3.3×10^{-6} mol/L) and the recoveries were in the range of 73.2–87.2 % and 90.5–103.7 % (spectrofluorometry). In another study, Effting et al. fabricated MMINS ($\text{Fe}_3\text{O}_4/\text{SiO}_2/\text{MIP}$) by means of a novel semi-covalent synthesis and successfully used it to pre-concentrate four TCs (chlor-tetracycline, doxycycline, OTC and TC) in water and milk samples prior to HPLC/DAD [37]. The prepared $\text{Fe}_3\text{O}_4/\text{SiO}_2/\text{MIP}$ showed high surface area ($216.0 \text{ m}^2/\text{g}$) and preconcentration factor (52.3–87.4).

Emerging various modification materials such as MOF [102], porous material [56,103], carbon dots [104,105], biochar [106], nanotubes [107], GO [108,109] and COFs [110,111] are rapidly finding a niche in the combination of MIPs to prepare magnetic composite adsorbents. For instance, Li and Wen employed Fe_3O_4 (core) and porous cellulose (carrier) to prepare magnetic porous cellulose MIPs (MPCMIP) [56]. Based on the unique characteristics of biodegradability, biocompatibility, convenience and environmental friendliness, the fabricated magnetic adsorbent presented satisfactory adsorption performance, adsorption rate and extraction selectivity. The imprinting factors towards TC, demeclocycline and minocycline were 1.49, 1.66 and 3.14, respectively. The detection of OTC based on fluorescence technique has been reported by Li [104]. In the work, a sensitive fluorescent probe based on boron-doped carbon dots (B-CDs) in the combination with thermo-responsive magnetic MIPs (T-MMIPs) was constructed for the detection of OTC in tea drinks. The proposed method integrated the merits of T-MMIPs and B-CDs, resulting in lower detection limit (0.1 $\mu\text{g}/\text{L}$) and higher selectivity. As a result, the integration of T-MMIPs and B-CDs owned capacious application prospect in the analysis of TCs in food samples.

3.4. β -lactams

As one of the most important groups of antibiotics, BLAs have been commonly used as antimicrobial drugs in the treatment of lactating dairy cattle for several contagious diseases for more than 80 years. However, these drug residues pose a threat to human health, such as allergic reactions in some hypersensitive individuals due to their carcinogenicity, teratogenicity and biotoxicity. Therefore, development of a rapid, selective and simple method for the quantification of BLAs has become a prerequisite for ensuring food safety and maintaining public health [117,118].

In this connection, Zhang et al. successfully prepared MMINS using penicillin V potassium (PENK) as template and MAA as functional monomer [119]. The obtained imprinting adsorbent was used to selectively enrich PENK, amoxicillin and oxacillin in milk. Combining with HPLC-MS/MS, sensitive method for the analysis of BLAs with satisfactory recovery (71.6–90.7 %) and sensitivity (LODs were in range of 1.6–2.8 $\mu\text{g}/\text{L}$) was established. Hao's group prepared MMINS and coupled with nano-spray ion source for the analysis of cephalosporin antibiotics in milk, egg and beef samples [120]. The established method showed high sensitivity with LODs ranged from 0.3 $\mu\text{g}/\text{kg}$ to 0.5 $\mu\text{g}/\text{kg}$, and the recoveries were 72.6–115.5 %. In addition, many studies have reported the application of MMINS for selective adsorption and removal of BLAs. Ma et al. synthesizes a novel type of magnetic MIPs (Fe_3O_4 -MER-MMIPs) with porous magnetic submicron materials as the carrier and TEOS as the functional monomer for the selective adsorption and removal of meropenem (MER) [121]. In the study, the preparation process was carried out in aqueous solvent without the employment of organic solvent. The resulting submicron materials possessed high adsorption capacity (11.49 mg/g), rapid adsorption kinetics (40 min), satisfactory selectivity towards MER (imprinting factor was 2.06), good applicability and excellent reusability (the adsorption efficiency remained above 94 % after eight cycles).

Sensors are an important tool in the determination of antibiotic residues. The MMINS-based sensor combines the advantage of MMINS and electrochemical analysis. Results have revealed that the MMINS-based electrochemical sensor and optical sensor were important way to analyze BLAs residues in complex matrices [65,68,122]. Lopez et al. synthesized an electrochemical sensors consisted of a carbon paste electrode (CPE) modified with magnetic MIP [65]. The results showed that the sensor has high sensitivity and selectivity, and could be used to detect amoxicillin and BLA antibiotics in real samples. The electrochemical sensor showed a linear ranged from 2.5 to 57 $\mu\text{mol}/\text{L}$ for amoxicillin ($R^2=0.9964$), the achieved LOD was 0.75 $\mu\text{mol}/\text{L}$. Chaitong et al. designed an optosensor-based MMINS probes to detect ultra-trace level of cefoperazone [68]. The sensor displayed good specificity for cefoperazone with an imprinting factor of 7.5. The MMINS-based probe successfully detected cefoperazone in milk and the spiked recoveries ranged from 85.3 % to 98.3 %.

3.5. Others

At present, more and more fast, effectively and sensitively analytical techniques based on MMINS have been developed to accurately detect trace antibiotic residues. Aside from commonly used SAs, QNs and TCs, other antibiotics such as MLs, BLAs, AGs, CAPs and LMs are also extensively employed to prevent and treat bacterial and parasitic infectious diseases in the livestock and poultry industry (Table 3).

MLs are lipophilic molecules with a central lactone ring of 12–16 carbon atoms to which several amino and/or neutral sugars are bound. MLs have become one of the most commonly used anti-infective drugs in clinical treatment due to their strong antibacterial activity against most bacteria and *Mycoplasma* [124]. However, the misuse of MLs may lead to antibiotic residues contamination in animal-derived foodstuff. Yang et al. presented a core-satellite structured magnetic nanosorbents ($\text{Fe}_3\text{O}_4@\text{PDA-HPMIP}$) and employed it as the adsorbent of MSPE for

selective extraction of MLs in honey sample. In combination with HPLC-MS/MS, a rapid and sensitive method for MLs analysis in honey was established. The developed method exhibited satisfactory linearity (0.4–40 µg/kg), ultra-high sensitivity (0.003–0.075 µg/kg) and precision (with RSD <4.3 %) [125].

AGs are an essential group of antibiotics composed of amino-modified sugars. As a type of broad-spectrum antibacterial activity, AGs have been widely used in veterinary medicines to treat infectious diseases for inhibiting the growth and reproduction of bacteria and promoting the growth of animals. However, these drug residues present serious toxic adverse effects on humans, and the maximum residue limits of AGs in food have been established by many countries to protect human health [69]. To analyze the trace AGs, Cao et al. successfully polymerized the dummy molecularly imprinted layer on the surface of Fe₃O₄ nanoparticles modified with NH₂ group [127]. The obtained magnetic adsorbent showed uniform sizes (~500 nm), highly magnetic responsivity (12.4 emu/g) and selective recognition capacity (20.5 mg/g) towards five AGs. Combining with HPLC-MS/MS, a method for enriching and determining AGs in milk samples with satisfactory recovery (82.6–114.1 %) and high sensitivity (3.6–9.6 µg/kg) was developed.

CAPs are highly effective broad-spectrum antibiotics that isolated from *Streptomyces Venezuela*. It has been widely used in the prevention and treatment of various diseases. However, it can accumulate in trace amounts in the aquatic environment and animals, resulting in toxic effects on humans. A number of countries around the world have made extremely stringent regulations on the use of CAPs [128]. At present, many studies have constructed highly sensitive and selective MMINs-based method for the determination of CAPs [129–132]. For example, Yao et al. constructed an electrochemical sensor (ZnWO₄@Co-MNPC@MIP/GCE) for the detection CAP in milk samples [129]. To develop the sensor, Co-MNPC nanomaterials were prepared by means of one-step hydrothermal technology and a direct carbonization method. And then, the ZnWO₄ was recombined with Co-MNPC and the MIP shells, followed by being coated on the surface of ZnWO₄@Co-MNPC to obtain ZnWO₄@Co-MNPC@MIP. After that, it was coated on the glassy carbon electrode. The constructed sensor showed excellent performance with a fairly wide linear range, high detection sensitivity (LOD was 0.0027 µmol/L) and satisfactory recovery (96.8 %–98.9 %). The study provided a practical sensing platform for the detection of CAP in milk.

As an important class of antibiotics, LMs are also widely applied in livestock farming. However, the LMs residues may lead to inefficient antibiotic therapeutics when the drug-resistant bacteria are produced. Therefore, it is urgent to develop simple and rapid methods to detect LMs in animal-derived foods. Cai et al. successfully constructed a core-shell structured MMIN through surface-initiated atom transfer radical polymerization on Fe₃O₄/PVBC/DVB with lincomycin hydrochloride (LIN) as template and 4-vinyl pyridine as functional monomer. Strong hydrophobic, π - π stacking and electrostatic interactions between LIN and MMIM contributed to the efficient adsorption. The prepared adsorbent presented good specific recognition ability towards LIN with the imprinting factor of 3.72. The LODs (15.0 ng/g) and satisfactory recovery (94.3–98.2 %) for milk revealed that the constructed method was reliable for determining LIN in real food samples [133].

4. Conclusions and future prospects

This review provides the latest progress of MMINs in the detection of antibiotics since 2010, mainly focused on the classification of MMINs, the strategies for synthesizing MMINs, and the typical applications of MMINs in the selective extraction of SAs, QNs, BLAs, TCs and other antibiotics. Although MSPE based on MMINs has achieved wide application in the analysis of antibiotic residues, some challenging issues still can be encountered and should be circumvented in the future study: (1) The concentrations of antibiotics are extremely low (nanogram per liter level) in complex samples. When these samples are treated with

synthesized MMINs, template, leakage may seriously affect the analytical accuracy and precision. Thereby, a dummy template imprinting strategy using structural analogues of the target compound as template needs further exploration. (2) So far, the reported MMINs-based sample pretreatment techniques mainly focus on the selective enrichment of single class of antibiotics, which is not conducive to achieving simultaneously selective enrichment of multi-classes targets. Hence, multi-templates imprinting strategy (MTIS) using multiple target molecules as templates should be developed. The MTIS can generate multiple types of recognition sites for selective extraction, isolation and detection of multiple classes of target species. (3) At present, the mainstream method for synthesizing MMINs-based adsorbent is the multi-step imprinting strategy, which requires tedious processing steps and consumes a large amount of organic reagents and preparation time. In this connection, green fabrication methodologies, from synthesis precursors to preparation conditions, should be paid for more particular attention. (4) Typically, the construction of the optimal imprinting system for MMINs requires repeated experiments to select the optimal functional monomer and cross-linker so as to produce the strongest interactions with the template. The procedure is time-consuming and laborious. Therefore, computer assistant technology, spectral prediction measure and characterization technique should be combined to quickly optimize the preparation conditions of MMINs with abundant recognition sites. (5) Great explorations are imperative to develop automatic on-line hyphenation of MMINs/MSPE with analytical instruments so as to enhance analytical efficiency. Last but not least, it is necessary to actively promote the large-scale production of MMINs-based adsorbents so as to push forward the development and wide application of MSPE technique in the isolation and enrichment of antibiotic residues and other pollutants in complex samples.

CRedit authorship contribution statement

Youfang Huang: Writing – original draft. **Cui Ma:** Project administration. **Xiaojia Huang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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